EE5203 L08 Worksheet

PWM, output compare, and input capture - Basri Erdogan

Expected time: 30-45 minutes

Policy: Attempt every question before consulting the worked solutions.

Timer clock = **16 MHz** throughout.

Part A - PWM arithmetic

1. Write the two formulas: PWM frequency in terms of `PSC` and `ARR`, and duty cycle in terms of `CCR` and `ARR`.
2. Complete the table. Show the arithmetic.

Requirement	`PSC`	`ARR`	`CCR`
1 kHz, 35 %			
2 kHz, 60 %			
200 Hz, 25 %			
50 Hz servo, 1.5 ms pulse			

3. Explain why `CCR = ARR` does not give 100 % duty. What value does it give for `ARR = 999`, and what would give a true 100 %?
4. A student wants to change the PWM *frequency* at run time and writes a new value into `CCR`. Describe what actually changes, and name the register they should have written.
5. TIM3 has four channels. A team wants a 1 kHz LED on CH1 and a 50 Hz servo on CH2, both on TIM3. Explain why this cannot work, and give the fix.
6. At `PSC = 15`, `ARR = 19999`, explain in one sentence why `CCR` can be read directly as "the pulse width in microseconds."

Part B - The handover

7. Name the register and bit-field that show a pin has been handed from GPIO output to a timer, and give the value it takes.
8. A student's PWM does nothing, but `HAL_GPIO_WritePin` still works on the same pin. Give the diagnosis and the one thing to check in the `.ioc`.
9. `HAL_TIM_PWM_Start()` is called but `CCER` bit 0 reads 0 afterwards. What would you suspect, and what is the visible symptom?

Part C - Capture mechanism

10. In one sentence each: what does `CC1S` in `CCMR1` select, and what does `CC1P` in `CCER` select in capture mode?
11. Who writes `CCR1` in capture mode, and at exactly what instant?
12. Explain **direct** and **indirect** capture mode. If CH1 is direct on PB6, what does CH2 listen to when configured indirect — and what would it listen to if configured direct instead?
13. Why are two channels needed to measure duty cycle, when one is enough to measure period?

Part D - Capture arithmetic

14. A 16-bit timer runs at a 1 µs tick. Rising captures arrive at `4000` and `9000`; the falling capture between them is `5250`. Compute the period, frequency, high time, and duty cycle.
15. Rising captures arrive at `64000` then `1500` on the same timer.
    a. Compute the period with the `uint16_t` cast.
    b. Compute what a plain `int` subtraction would give.
    c. Explain the difference in terms of modular arithmetic.
16. What is the longest interval a 16-bit timer at a 1 µs tick can measure honestly? Give the answer in milliseconds and as a signal frequency.

17. A team measures a 10 Hz signal with that setup and gets a stable, plausible reading of about 35 ms. Explain what went wrong and give **two** different fixes.

18. Your capture code works on TIM4 (16-bit). The team moves it to TIM2 (32-bit) unchanged, keeping the `(uint16_t)` casts. What now goes wrong, and when?

Part E - Evidence and behaviour

19. Fill in the evidence you would show for each claim.

Claim	SFR observation
The duty is the value I configured	
The pin belongs to the timer	
Hardware is timestamping without my code	
The capture edge is the one I selected	
A capture was lost	

20. Describe an experiment, using only the debugger and the SFR view, that makes `CC1OF` become set on purpose. Explain what was lost.

21. You flip `CC1P` for the capture channel while the program runs. Predict what happens to the measured **period**, and to the measured **duty**. Justify both answers.

22. The generating side changes its duty with B1 (one `CCR1` write per press) while the measuring side reports duty. How soon after a press does the reported duty show the new level? What would happen if the generator's `ARR` changed at the same moment?

Exit self-assessment

☐ I can derive `PSC`/`ARR`/`CCR` for any frequency and duty.
☐ I can explain the 100 % off-by-one without looking it up.
☐ I can say who writes `CCR` in each direction, and when.
☐ I can compute period and duty from captures, wrap-safe.
☐ I can state the honest measurement range of a 16-bit capture timer.